

Geometry

Unit 8

Lesson 6 Homework

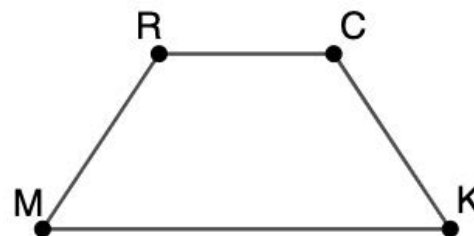
Name _____

Date _____

Period _____

Two congruent angles isosceles trapezoid $KMRC$ have measures $(3x + 10)^\circ$ and $(5x - 10)^\circ$.

1. What is the value of x ?



2. What are the measures of all angles of the trapezoid?

For each quadrilateral description, select all shapes that match the description.

3. Quadrilateral $GRMW$ has two pairs of parallel sides and diagonals are congruent.

- ☐ Parallelogram
- ☐ Rectangle
- ☐ Square
- ☐ Rhombus
- ☐ Kite

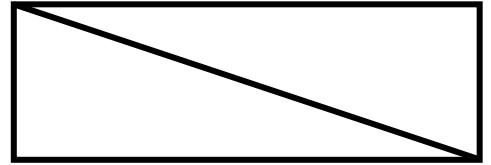
4. Quadrilateral $BNTV$ has diagonals are perpendicular to each other and two pairs of parallel sides.

- ☐ Parallelogram
- ☐ Rectangle
- ☐ Square
- ☐ Rhombus
- ☐ Kite

5. Quadrilateral $KPSX$ has two pairs of congruent sides and diagonals are perpendicular to each other.

- ☐ Parallelogram
- ☐ Rectangle
- ☐ Square
- ☐ Rhombus
- ☐ Kite

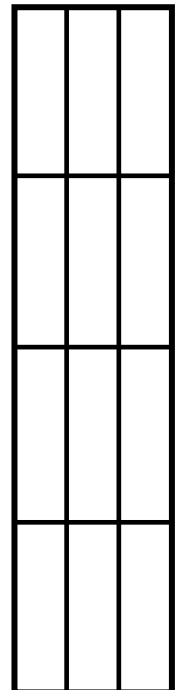
Barrett is looking at a building on the school grounds which is shaped like a rectangle as shown. He knows the building has a height of $25x$ feet and a width of $28x$ feet with a diagonal of 180 feet.



6. What is the value of x ? Round your answer to the nearest hundredth.
7. What is the height of the building? Round your answer to the nearest hundredth.
8. What is the width of the building? Round your answer to the nearest hundredth.

Use the given figure to answer questions 9-10.

A wall has paneling made up of rectangles, with an area of 30 square feet each, as shown. There are four rectangles along the height and three along the top. The edge of each rectangle in the paneling is $(2x + 3)$ feet in width and $(4x + 2)$ feet in height.



9. What is the value of x ?
10. What are the dimensions of the portion of the wall covered the paneling?